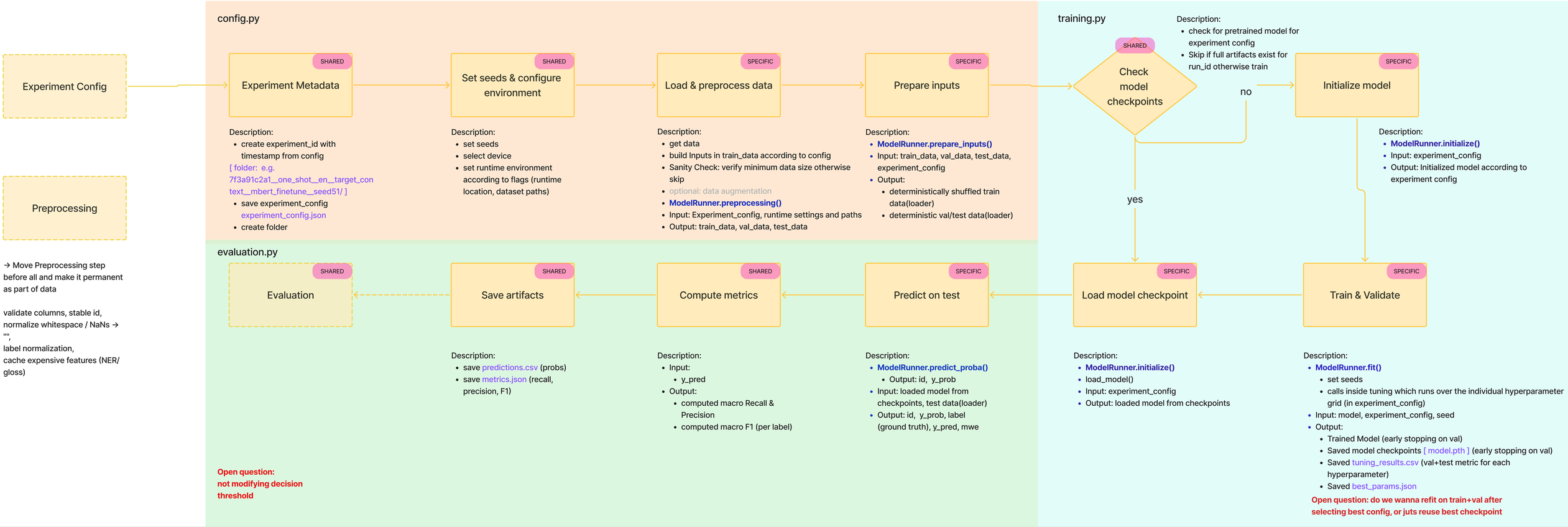
Experiment Config
Preprocessing
→ Move Preprocessing step before all and make it permanent as part of data
validate columns, stable id, normalize whitespace / NaNs → "",
label normalization,
cache expensive features (NER/ gloss)
config.py
SHARED
Experiment Metadata
Description:
• create experiment_id with timestamp from config
[folder: e.g. 7f3a91c2a1__one_shot__en__target_context__mbert_finetune__seed51/]
• save experiment_config experiment_config.json
• create folder
SHARED
Set seeds & configure environment
Description:
• set seeds
• select device
• set runtime environment according to flags (runtime location, dataset paths)
SPECIFIC
Load & preprocess data
Description:
• get data
• build Inputs in train_data according to config
• Sanity Check: verify minimum data size otherwise skip
• optional: data augmentation
• ModelRunner.preprocessing()
• Input: Experiment_config, runtime settings and paths
• Output: train_data, val_data, test_data
SPECIFIC
Prepare inputs
Description:
• ModelRunner.prepare_inputs()
• Input: train_data, val_data, test_data, experiment_config
• Output:
• deterministically shuffled train data(loader)
• deterministic val/test data(loader)
training.py
SHARED
Check model checkpoints
Description:
• check for pretrained model for experiment config
• Skip if full artifacts exist for run_id otherwise train
no
yes
SPECIFIC
Initialize model
Description:
• ModelRunner.initialize()
• Input: experiment_config
• Output: Initialized model according to experiment config
SPECIFIC
Train & Validate
Description:
• ModelRunner.fit()
• set seeds
• calls inside tuning which runs over the individual hyperparameter grid (in experiment_config)
• Input: model, experiment_config, seed
• Output:
• Trained Model (early stopping on val)
• Saved model checkpoints [model.pth] (early stopping on val)
• Saved tuning_results.csv (val+test metric for each hyperparameter)
• Saved best_params.json
Open question: do we wanna refit on train+val after selecting best config, or juts reuse best checkpoint
SPECIFIC
Load model checkpoint
Description:
• ModelRunner.initialize()
• load_model()
• Input: experiment_config
• Output: loaded model from checkpoints
evaluation.py
SPECIFIC
Predict on test
Description:
• ModelRunner.predict_proba()
• Output: id, y_prob
• Input: loaded model from checkpoints, test data(loader)
• Output: id, y_prob, label (ground truth), y_pred, mwe
SHARED
Compute metrics
Description:
• Input:
• y_pred
• Output:
• computed macro Recall & Precision
• computed macro F1 (per label)
SHARED
Save artifacts
Description:
• save predictions.csv (probs)
• save metrics.json (recall, precision, F1)
SHARED
Evaluation
Open question:
not modifying decision threshold